UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

# Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 001 · Introduction to partial differentiation · compiled 15 October 2026

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|---|---|
| ESA ★★★ | Asked repeatedly in past ESA papers |
| ESA ★★ | Asked in past ESA papers or computed in class-work problems |
| ESA ★ | Conceptual: taught in class, not examined so far |
| ADDED FOR CLARITY | Standard context that is not in the class material |
| TRANSCRIPT ONLY | Said in the lecture but not on any slide |
| VIDEO ONLY | Seen only in a frame of the class video |
| NOT ON CURRENT SLIDES | From older material or past papers, not this year's slides |
| S001·15 | Deck 001, slide 15 (rendered .pptx) |
| T001a 00:14:20 | Transcript “Introduction to Partial Differentiation”, segment start |
| T001b 00:11:41 | Transcript “Continuity and differentiability of a function of two independent variables” |
| CW U1 P7 | Unit 1 class-work problem 7 |
| ESA Dec-23 1a | Past paper: Dec 2023, question 1.a |

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4. Figure G2.2 · $\frac{2xy}{3x^2+y^2}$ along the line $y = mx$
5. Figure G3.1 · $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists
6. Figure G3.2 · $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist


Colours follow the Okabe–Ito order: blue = main surface or curve, vermilion = comparison, green and purple = further paths. Figures G1.1–G1.2 and G3.1–G3.2 redraw the deck's Gnuplot surfaces ($\sin x \cos y$, $e^{1-x^2-y^2}$, $\frac{3x^2y}{x^2+y^2}$, $\frac{2xy}{3x^2+y^2}$) over $-2 \le x, y \le 2$ or a similar range; the slides' screenshots are not copied. Figure G2.1 redraws the lecture's board sketch of paths approaching $(a, b)$.

# 1 Functions of two or more variables

$z = \sin x \cos y$

$-\pi \le x,\ y \le \pi;\ \ -1 \le z \le 1$

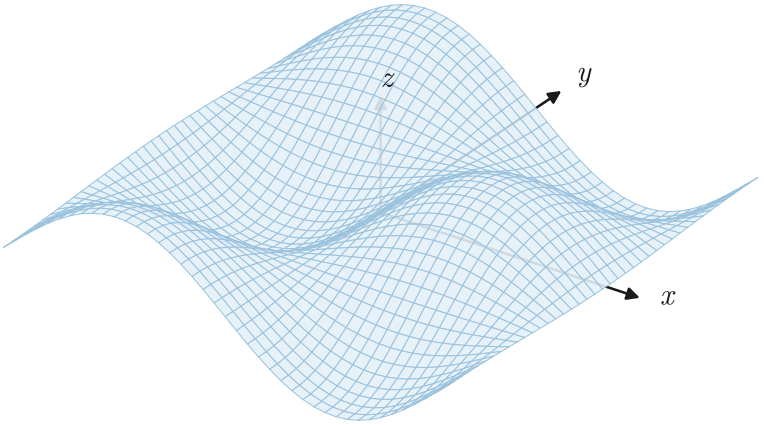



**Figure G1.1.** Graph of $z = \sin x \cos y$. ESA ★

S001·8 · T001a 00:09:07

***How to read it.*** A function of two variables is a surface over the $xy$-plane: above each point $(x, y)$ the height is $f(x, y)$. $\sin x \cos y$ is an egg-box of hills and hollows between $-1$ and $1$; the slide's Gnuplot picture shows the same surface.

$z = e^{1 - x^2 - y^2}$

top $(0, 0, e)$

green: level curves $z =$ const (circles)

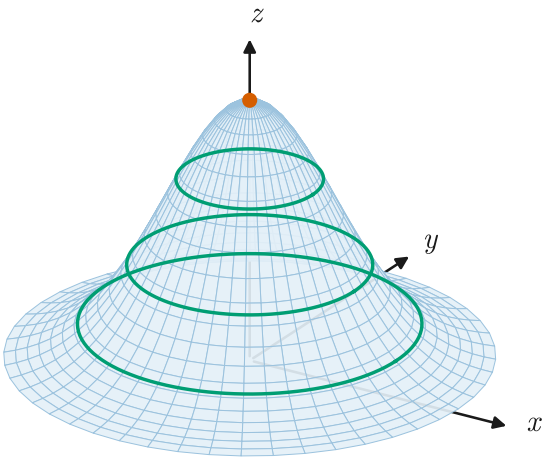



**Figure G1.2.** Graph of $z = e^{1-x^2-y^2}$. ESA ★

S001·8 · T001a 00:09:07

***How to read it.*** A single bell-shaped hill with its top $z = e$ above the origin, falling to 0 in every direction. The green level curves (drawn on the surface) are circles $x^2 + y^2 =$ constant.

# 2 Limit of a function of two variables

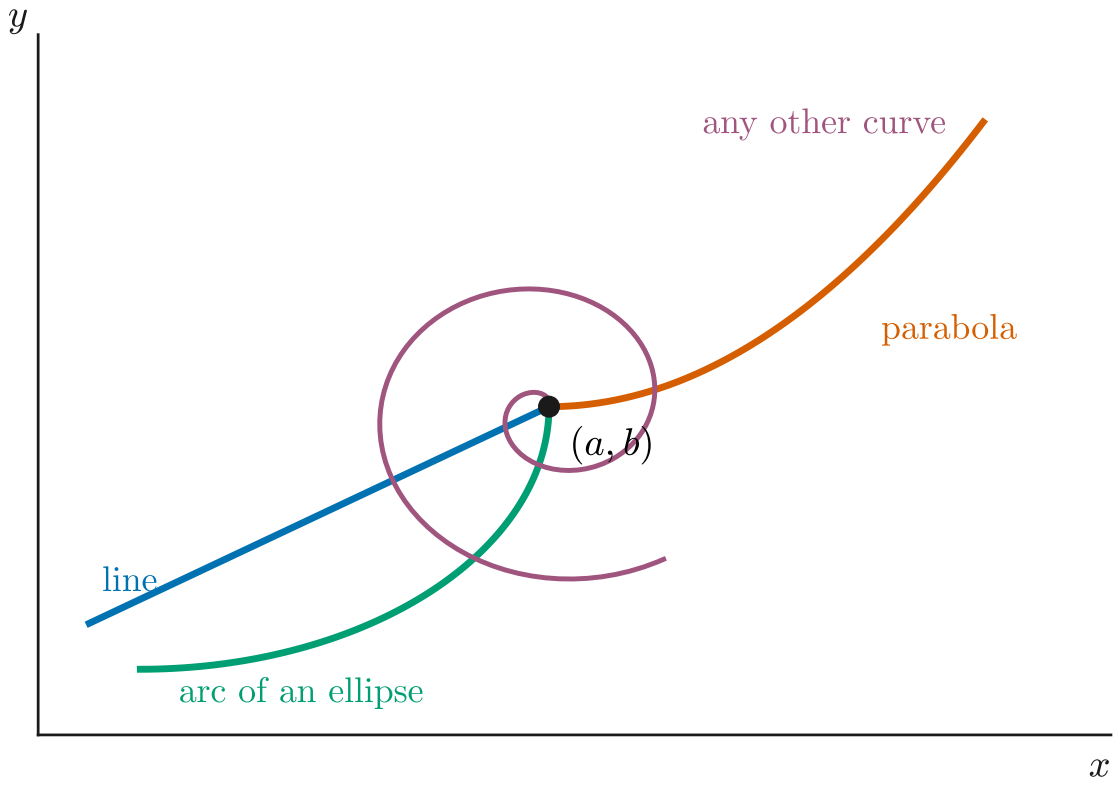



**Figure G2.1.** Paths along which $(x, y)$ can approach $(a, b)$. ESA ★

S001·9 · T001a 00:11:52 · T001a 00:12:19

***How to read it.*** In one variable $x$ reaches $a$ only from the left or the right. In the plane, $(x, y)$ can reach $(a, b)$ along a line, a parabola, an arc of an ellipse, or any other path. The limit exists only if all of them give the same value.

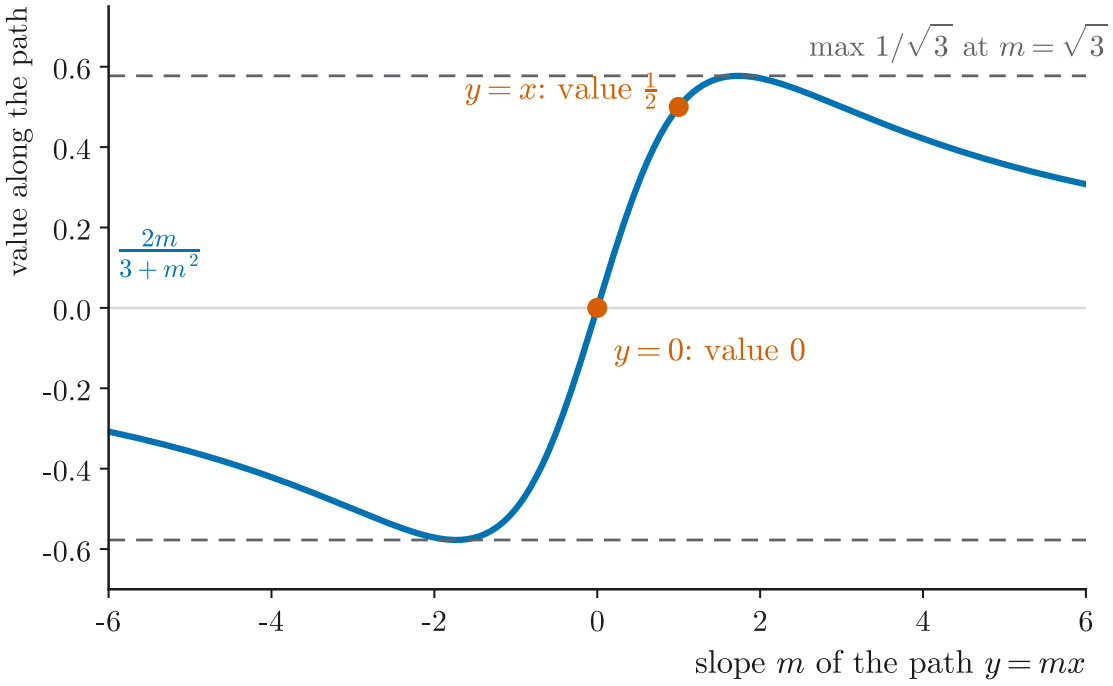



**Figure G2.2.** $\frac{2xy}{3x^2+y^2}$ along the line $y=mx$. ESA ★★

ADDED FOR CLARITY S001·10 · T001a 00:17:24

***How to read it.*** The value of $\frac{2xy}{3x^2+y^2}$ on the line $y=mx$ is the constant $\frac{2m}{3+m^2}$, plotted against the slope $m$. $m=0$ gives 0, $m=1$ gives $\frac{1}{2}$, and the largest value is $\frac{1}{\sqrt{3}}$ at $m=\sqrt{3}$. Because the value changes with the path, the limit at the origin does not exist.

# 3 Continuity of a function of two variables

$$z = \frac{3x^2 y}{x^2 + y^2}$$

limit 0 at the origin; $f(0,0)$ undefined

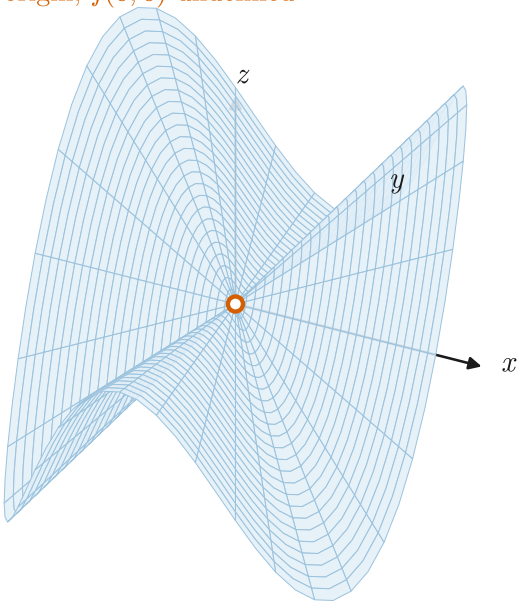



**Figure G3.1.** $z = \frac{3x^2y}{x^2+y^2}$: the limit at the origin exists. ESA ★

S001·12 · T001b 00:02:41

***How to read it.*** The surface folds near the origin but every direction leads to height 0 there (vermilion point). It is missing only the single point $(0,0)$, where $f$ is undefined: a removable discontinuity.

$z = \dfrac{2xy}{3x^2 + y^2}$

ray $y = x$: height $\frac{1}{2}$

$x$-axis: height 0

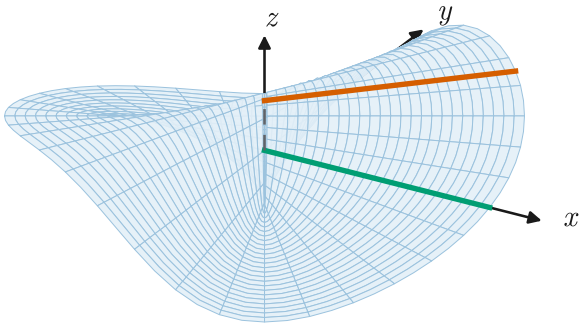



**Figure G3.2.** $z = \frac{2xy}{3x^2+y^2}$: the limit at the origin does not exist. ESA ★

S001·13 · T001b 00:06:13

***How to read it.*** Each ray from the origin keeps a constant height $\frac{2m}{3+m^2}$, so the surface is a twisted fan of straight lines meeting at the $z$-axis at different heights (vermilion: the ray $y = x$ at height $\frac{1}{2}$; blue: the $x$-axis at height 0). There is no single value at the origin.